



C-language Portfolio

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Definition: C language ka aik buniyadi program structure hota hai jismein main() function se execution start hoti hai. Coding scheme ka matlab code ko saaf-suthra, indentation ke saath aur rules ke mutabiq likhna hai taake compiler aur parhne wale ko samajh aaye.

C

```
#include <stdio.h>

int main() {
    printf("Welcome to my C Portfolio!\n");
    return 0;
}
```

Definition: Data types compiler ko batati hain ke variable kis qisam ka data store karega, jaise integer (int), decimal (float), ya single character (char).

C

```
#include <stdio.h>

int main() {
    int age = 18;
    float gpa = 3.5;
    char grade = 'A';

    printf("Age: %d\n", age);
    printf("GPA: %.2f\n", gpa);
    printf("Grade: %c\n", grade);
    return 0;
}
```

Definition: Program flow ka matlab yeh hai ke code kis tarah se run hoga. Control structures (jaise if-else ya switch-case) condition ke mutabiq faisla karte hain ke kaunsa block execute hoga.

C

```
#include <stdio.h>

int main() {
    int marks = 75;
    if (marks >= 50) {
        printf("Pass ho gaye!\n");
    } else {
        printf("Fail ho gaye.\n");
    }
    return 0;
}
```

Definition: Loop ka istemal karke ek khas tarteeb (sequence) ya pattern ke mutabiq numbers generate karna series kehlata hai (jaise even numbers ki series).

C

```
#include <stdio.h>

int main() {
    int i;
    printf("Even numbers series (1 to 10): ");
    for (i = 1; i <= 10; i++) {
        if (i % 2 == 0) {
            printf("%d ", i);
        }
    }
    return 0;
}
```

Definition: Jab hamein pata ho ke aik hi kaam ko kitni dafa repeat karna hai, toh hum for loop ka istemal karte hain. Isme initialization, condition, aur increment/decrement aik hi line mein hote hain.

C

```
#include <stdio.h>

int main() {
    int i;
    for (i = 1; i <= 5; i++) {
        printf("Number: %d\n", i);
    }
    return 0;
}
```

Definition: Yeh loop tab use hota hai jab pehle se iterations ki tadad fix na ho, balki sirf aik condition check karni ho ke jab tak condition true hai, loop chalta rahe.

C

```
#include <stdio.h>

int main() {
    int count = 3;
    while (count > 0) {
        printf("Countdown: %d\n", count);
        count--;
    }
    return 0;
}
```

Definition: Yeh while loop ki tarah hi hai, lekin iska yeh faida hai ke condition baad mein check hoti hai, is liye loop ka block **kam az kam aik dafa zaroor** run hota hai chahe condition false hi kyun na ho.

C

```
#include <stdio.h>

int main() {
    int num = 1;
    do {
        printf("Yeh aik dafa zaroor chalega. Value: %d\n", num);
        num++;
    } while (num < 1);
    return 0;
}
```

Definition: break statement loop ko darmiyan mein hi rok deti hai, jabke continue statement current iteration ko skip karke agli iteration par chali jati hai.

C

```
#include <stdio.h>

int main() {
    int i;
    for (i = 1; i <= 5; i++) {
        if (i == 3) {
            continue; // 3 ko skip kar dega
        }
        if (i == 5) {
            break;    // 5 par loop rokh dega
        }
        printf("%d ", i);
    }
    return 0;
}
```

Definition: Aik loop ke andar doosra loop lagane ko nested loop kehte hain. Yeh aam tor par patterns (jaise stars ya matrix) print karne ke liye istemal hota hai.

C

```
#include <stdio.h>

int main() {
    int i, j;
    for (i = 1; i <= 3; i++) {
        for (j = 1; j <= i; j++) {
            printf("* ");
        }
        printf("\n");
    }
    return 0;
}
```

Definition: Array aik aisi variable hoti hai jo aik hi waqt mein aik hi data type ki bohot saari values ko consecutive memory locations par store kar sakti hai.

C

```
#include <stdio.h>

int main() {
    int marks[3] = {80, 85, 90};
    int i;
    for (i = 0; i < 3; i++) {
        printf("Subject %d marks: %d\n", i + 1, marks[i]);
    }
    return 0;
}
```